Site Acceptance Test

Revisions

2	26/10/2021	R&D Manager/ A.Nicchio	QA/HSEM- C.Delli Quadri/M. Tremolada	Engineering & Product Director/ R.Ciceri	Added Chapter 2 “Acronyms”, Modified par.2.2 (now 3.2) and 2.3 (now 3.3), format revision and update.
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1. Scope

The scope of this procedure is to indicate how mechanical running test has to be carried out to verify the good and correct working of pumps manufactured by Trillium Pumps Italy.

Additionaly, pump developed head and other performance parameters can be measured to verify consistency with FAT values, but verification and acceptance of pump performances shall be anyway based only on functional tests (Performance, Mechanical run and NPSH tests as required by job specification) held at manufacturer facility. In facts, operational constraints, piping layout, fluid type do not allow measurement at site to be in compliance with international standards that regulate pump performance acceptability. Consistency of performance at site with FAT shall be based on sound engineering practice and personnel expertise so that eventual variation can be evaluated on a case by case basis.

Hydraulic performances acceptability criteria and NPSHr test are excluded from the scope of Site Operation Assessment SAT and so from the contents of this procedure.

IMPORTANT: **Safety warnings are not included in this document, please refer to IOM manuals. Every performed activity shall be in agreement with relevant Safety procedures and local regulation. TPI shall not be held responsible for any safety related issue derived by the use of this procedure.**

2. Acronyms

- **FAT**: Factory Acceptance Test
- **NPSH**: Net Positive Suction Head
- **RMS**: Root Mean Square
- **RTD**: Resistance Temperature Detector
- **SAT**: Site Acceptance Test
- **TPI**: Trillium Pumps Italy

3. Mechanical Acceptance Criteria

The below paragraphs refer to the bare shaft pump only. The driver and all other eventual train components such as gearboxes shall be evaluated. Refer to specific manuals of each equipment for limits and acceptance criteria.

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3.1. Bearing temperature

Bearing temperature shall be measured during SAT and shall be found within the below limits. Limits are not absolute values but shall be referred to ambient temperature, which shall be measured.

Conditions of measurement	Maximum allowed temperature increase
Antifriction Bearing outer race / bearing shoe (when RTD are installed)	+ 50°C
Bearing oil bath (oil ring/splash systems)	+ 39°C
Bearing housing outer surfaces	+39°C

When RTD are not installed, a portable RTD on external bearing housing surfaces shall be used. This instrument shall be calibrated, and the calibration certificate shall be valid at the date of measurement.

The same portable RTD shall be used to measure the ambient temperature close to the pump.

Measure shall be repeated every 30 min for 4 hours, at rated flow (refer to pump datasheet for each item specific values).

3.2. Bearing and Shaft Vibrations

Vibration will be checked and measured according to actual flow.

The following tables should be adopted as limitations on the field vibration values, based upon the pump type (horizontal / vertical):

Horizontal pumps		
Type of measure	**Acceptability limit within preferred region**	**Acceptability limit within allowable region**
Accelerometer	Overall 3.5 mm/s RMS	Overall 4.6 mm/s RMS
Proximity probes	50 μm pk/pk	65 μm pk/pk

Vertical pumps		
Type of measure	**Acceptability limit within preferred region**	**Acceptability limit within allowable region**
Accelerometer	Overall 5.5 mm/s RMS	Overall 7.2 mm/s RMS

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The above limits shall be increased according to API 610 XI Ed. Figure 34 for pumps absorbing more than 300 kW per stage or running at speed greater than 3600 rpm.

All vibration values shall be measured in the range 5÷1000 Hz.

In case overall values are exceeded, spectrum analysis is recommended to understand high vibration cause and determine corrective actions.

Limitations on baseplate/and soleplate vibration shall also be considered in judging vibration levels (see paragraph 3.3).

3.3. Baseplate / Soleplate vibration

The above-mentioned limits do not take into account the influence of external vibration on pump skid.

For horizontal pumps, once installed at site, both grouted and ungrouted baseplate shall not allow vibration in vertical direction (measured at lowest point on pump pedestal) greater than 0.7 mm/s RMS with pump in operation and 0.5 mm/s RMS when the pump is stopped.

For vertical pumps when an excessive background vibration is suspected, measurements should be made with the machine shut down, to determine the degree of external influence. If the vibration with the stationary machine exceeds 25 % of the value measured when the machine is running, corrective action is recommended, to reduce the effect of background vibration.

Deviation from the above reference values shall be investigated, values judged by properly trained personnel and eventually rectified (e.g.: re-fill of ungrouted areas).

4. Performance Assessment

During SAT all main process values as well as vibration and temperature levels shall be recorded and verified against approved datasheets, applicable project specification and standards.

It has to be noted that light discrepancies between results obtained at site and the ones recorded at manufacturer's shop could occur.

Differences between site and manufacturer's plants in terms of foundation, routing of piping and installed instruments and devices have direct effect on pump performance and behavior.

Tolerances on pump performances at site are therefore not predictable.

Acceptance of the provided skid shall be anyway based on results obtained during factory acceptance test held at the presence of Contractor's and Client's representative.

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Only pump performance test certificate shall be used as a reference while determining pump operating point with Head vs Flow curve.

4.1. Head measurement (informative)

Total Differential head ΔH can provide a clear indication of proper pump operation. Manometers at both suction and discharge pipes are required to perform measurements; furthermore, actual specific gravity at actual temperature shall be known. Head measurement is also useful to deduce the pump flowrate from the pump performance test certificate.

The total differential head can be calculated according to the following:

$$\Delta \mathrm{H} = \frac{(P_{disch} - P_{suct}) * 100000}{\rho \cdot g} + (Z_d - Z_s) + \Delta H_{kin}$$

Where:

$P_{disch} = pressure\ at\ discharge\ (before\ any\ valve, filter \ldots etc)\ in\ bar(g)$

$P_{suct} = pressure\ at\ suction\ (before\ pump\ inlet, after\ eventual\ filter)\ in\ bar(g)$

$\rho = fluid\ density\ in \frac{kg}{m^3}.\ This\ shall\ be\ verified\ with\ proper\ laboratory\ measures\ and\ referred\ to$

$actual\ temperature\ of\ pumped\ fluid.$

$g = free\ fall\ accelaration\ 9.81\ m/s^2$

$Z_d = elevation\ of\ discharge\ manometer\ respect\ to\ impeller\ Center\ Line\ in\ meters$

$Z_s = elevation\ of\ suction\ manometer\ respect\ to\ impeller\ Center\ Line\ in\ meters.$

$$\Delta H_{kin} = \frac{(V_d^2 - V_s^2)}{2g}$$

$V_d = fluid\ velocity\ in\ discharge\ pipe\ [\frac{m}{s}]$

$V_s = fluid\ velocity\ in\ suction\ pipe\ [\frac{m}{s}]$

Please note that in case flowrate cannot be measured, kinetic term cannot be calculated; in this case, its contribution to total head shall be neglected (with consequent decrease of accuracy of head measurement).

Furthermore, this value can be directly compared with test values only for non-viscous fluids. In case viscosity effect are relevant, data cannot be directly compared.

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4.2. Rotational speed measurement (informative)

Rotational speed can be easily measured by an optical tachometer. Use a reflective target on rotor may be required to properly perform reading.

4.3. Power Measurement (informative)

Measure of absorbed power is also useful to determine pump operating conditions. In order to measure electric values, the following steps shall be followed:

1) Measure Ampere I on each phase; values should be strictly similar (refer to electric motor documentation). Differences greater than 5 % should be investigated and rectified.
   Arithmetic average of I shall be used for power calculation.
2) Measure Voltage V input on each phase; values should be strictly similar (refer to electric motor documentation). Differences greater than 5 % should be investigated and rectified. Voltage readings should be performed at motor junction box to account for voltage losses due to cables resistance (especially for low voltage, long cables).
3) Arithmetic average of V shall be used for power calculation
4) Check datasheet for power factor (pf) and motor efficiency (η_{mot}).
   Pf and motor efficiency can be read from motor performance curve as a function of motor slip "$S_{\%}$". The value of slip "$S_{\%}$" shall be calculated to read power factor and motor efficiency with the following formulae:

$$S_{\%} = \frac{\omega_0 - \omega_{act}}{\omega_0} \cdot 100$$

$$\omega_0 = \frac{f * 60}{np/2}$$

   Where *f* is electric frequency, *np* number of poles.
   Alternatively, nominal values could be used, but this may result in higher uncertainties.
5) Calculate pump absorbed power with the following formula:

$$\mathrm{P_{abs}} = \sqrt{3} \cdot \mathrm{V} \cdot \mathrm{I} \cdot \mathrm{pf} \cdot \eta_{mot}$$

Please note that power is also sensitive to viscosity values, but these cannot be easily considered; comparison of power values to rated ones implies that there are no differences in liquid viscosity.

5. Table for performance record

Table reported here below should be used as template to collect data. Suggested time step for measurement is 30 min.

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Site Acceptance Test

The measurement report shall be signed by all parties involved in pump site acceptance.

5.1. OH pump type

Date/Time	Flow [m3/h]	Head [m]	rpm [rpm]	VDE OA [mm/s RMS]	HDE OA [mm/s RMS]	AX OA [mm/s RMS]	Bearing Temp [°C]	Ambient Temp [°C]

5.2. BB pump type (Ball - Ball bearing housing equipped)

Date	Flow [m3/h]	Head [m]	Rpm [rpm]	VDE OA [mm/s RMS]	HDE OA [mm/s RMS]	DE temp [°C]	VNDE OA [mm/s RMS]	HNDE OA [mm/s RMS]	AXNDE OA [mm/s RMS]	NDE temp [°C]	Ambient temp [°C]

5.3. BB pump type (Sleeve - Ball bearing housing equipped)

Date	Flow [m3/h]	Head [m]	Rpm [rpm]	X DE OA [µm pk/pk]	Y DE OA [µm pk/pk]	DE temp [°C]	X NDE OA [µm pk/pk]	Y NDE OA [µm pk/pk]	AXNDE OA [mm/s RMS]	VDE OA [mm/s RMS]	NDE temp [°C]	Ambient temp [°C]

5.4. BB pump type (Sleeve - Sleeve Bearings equipped)

Date	Flow [m3/h]	Head [m]	Rpm [rpm]	X DE OA [µm pk/pk]	Y DE OA [µm pk/pk]	DE* temp [°C]	X NDE OA [µm pk/pk]	Y NDE OA [µm pk/pk]	AXNDE OA [µm pk/pk]	NDE* radial temp [°C]	NDE* thrust temp [°C]	Ambient temp [°C]

5.5. Vertically Suspended pumps (all)

Date/time	Flow [m3/h]	Head [m]	rpm [rpm]	BH X direction O.A. [mm/s RMS]	BH Y direction O.A. [mm/s RMS]	BH AX O.A. [mm/s RMS]	Bearing housing Temp [°C]	Motor flange X direction O.A. [mm/s RMS]	Motor flange Y direction O.A. [mm/s RMS]	Ambient temp. [°C]

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6. References

- ISO 10816-7, Mechanical Vibration- Evaluation of machine vibration by measurement on non-rotating part.
- ISO 10816-3, Mechanical vibration — Evaluation of machine vibration by measurements on non-rotating parts — Part 3: Industrial machines with nominal power above 15 kW and nominal speeds between 120 r/min and 15 000 r/min when measured in situ.
- ISO 13709:2009 Centrifugal pumps for petroleum, petrochemical, and natural gas industries.
- API 610 XI Ed.

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